Performance and cost experiments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Performance and cost experiments before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Accessibility and user feedback

Without opening the old lesson, explain one key idea from Accessibility and user feedback. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Incident investigation

Without opening the old lesson, explain one key idea from Incident investigation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Deployment environments

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Performance experiments compare **matched workload**s with declared timing boundaries and quality constraints. Averages can hide slow cases.

Use the same cases and environment; record every duration and success. For a simple **median** sort values and average the middle pair for even counts. Keep failed requests visible.

Practise with fading help

> Complete the supported Performance and cost experiments trace and record the first step you can explain without help.

Use W2: Trace the worked case for Performance and cost experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Performance and cost experiments explanation against the supplied facts before discussing it.

Use the worked case for Performance and cost experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Performance and cost experiments facts and

> keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Performance and cost experiments facts and keep your reasoning.

Durations A=[1, 1, 1, 9] seconds, B=[2, 2, 2, 2]. All pass. Compute mean and **median**. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Performance and cost experiments facts and keep your reasoning.

A report omits this qualification: Resource improvement under one workload is not universal efficiency or equal quality. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Performance and cost experiments facts and keep your reasoning.

C=[1, 2, 3, 10], D=[3, 3, 3, 3]. Compute means/**median**s; one D response is wrong.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Performance and cost experiments facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Performance and cost experiments to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled performance and cost experiments **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of performance and cost experiments.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

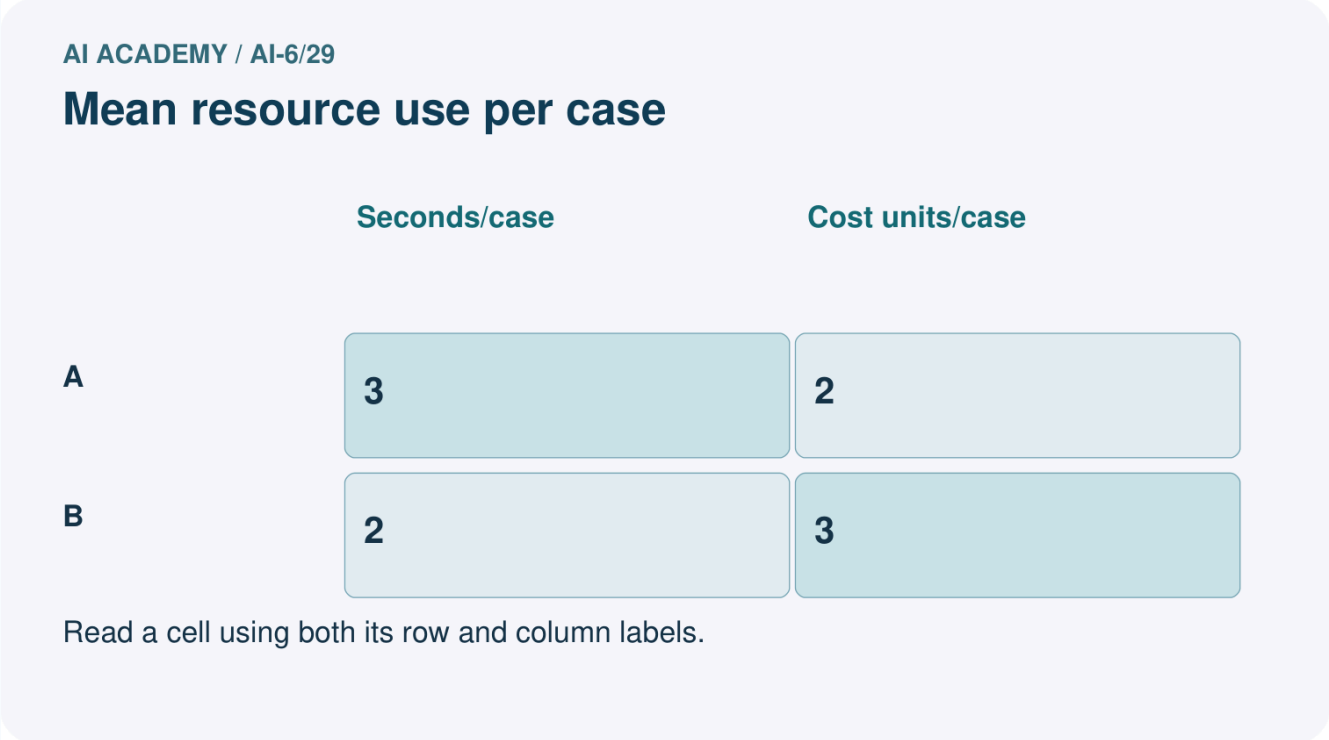
Pictorial case practice: Performance and cost experiments

> Measure time and cost. Two fixed configurations answer the same six questions and meet the

> same quality rubric. Explain the labelled step before checking the changed condition.

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/29

Performance and cost experiments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute mean time

A averages 3 seconds; B averages 2 seconds per case.



2. Compute mean cost

A averages 2 units; B averages 3 units per case.



3. Compare trade-off

B saves one second but costs one extra unit per case.

CHANGE ONE THING

Require at most 2 cost units per case.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate per-case seconds and cost units for A and B. Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B saves one second but costs one extra unit per case.

1. Compute mean cost

2. Compute mean time

3. Compare trade-off

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare trade-off B saves one second but costs one extra unit per case. A fair performance experiment holds the workload and measurement procedure fixed.

Faster is not automatically preferable under a budget.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Require at most 2 cost units per case. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. A meets the budget; B does not.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A meets the budget; B does not. The mandatory resource condition changes the decision.

Repair a missing building block

Review Accessibility and user feedback (ai-6/28). Goal: Explain the **mechanism** and terminology of accessibility and user feedback.

Short practice: Explain what each text, keyboard and assistive-technology change adds to the feedback. Why does changing the red circle to orange alone not solve the communication barrier?.

Return to this week and explain what you changed.

Review Incident investigation (ai-6/26). Goal: Explain the **mechanism** and terminology of incident investigation. Short practice: Order the three incident records and locate the old **evidence** version.

How does the investigation change if the trace instead retrieves version 6 but drafts room C?. Return to this week and explain what you changed.

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

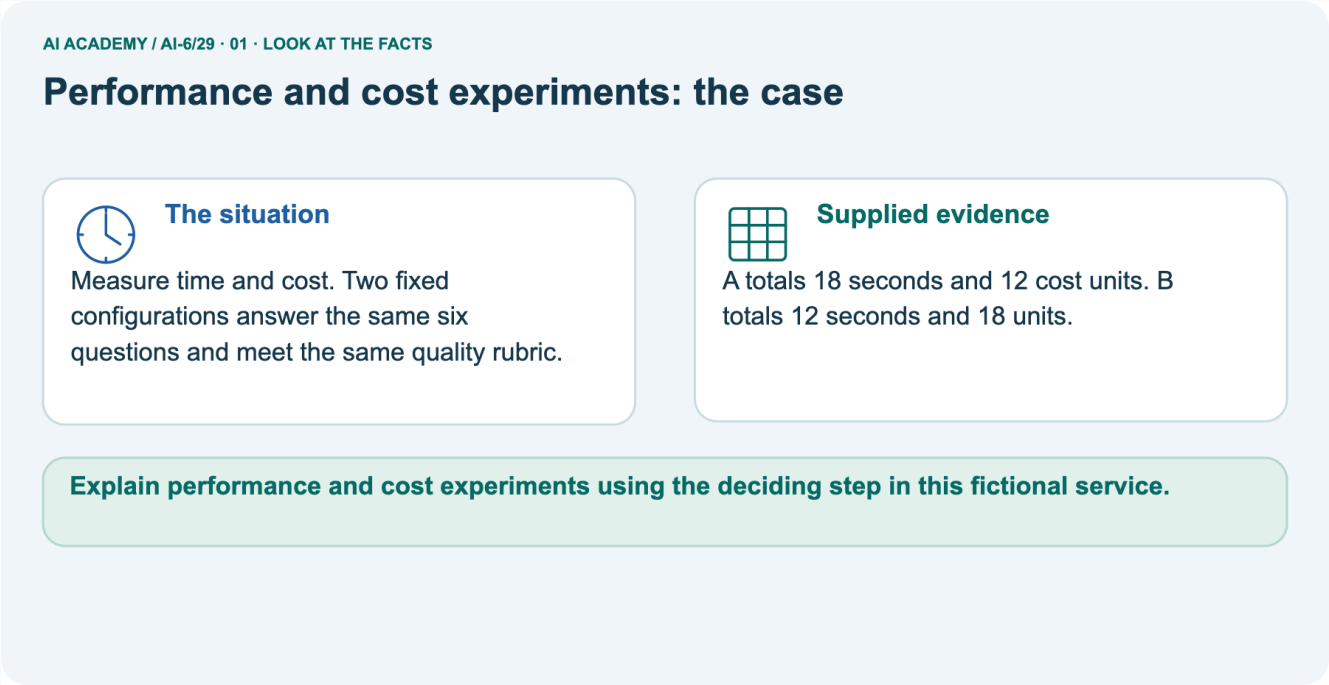
> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Performance and cost experiments picture from its labels, then record a relevant limit.

01. Performance and cost experiments: the case



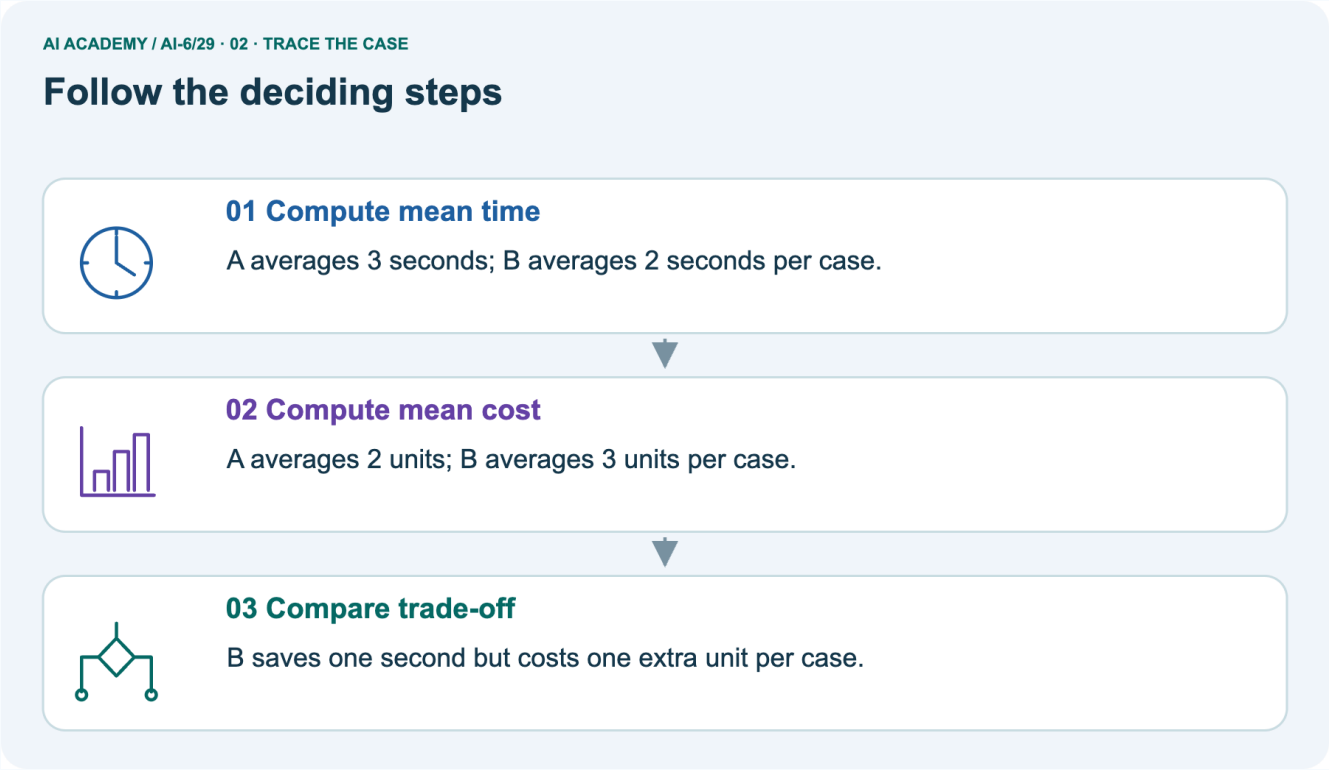
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure time and cost. Two fixed configurations answer the same six questions and meet the same quality rubric. Supplied facts: A totals 18 seconds and 12 cost units. B totals 12 seconds and 18 units.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



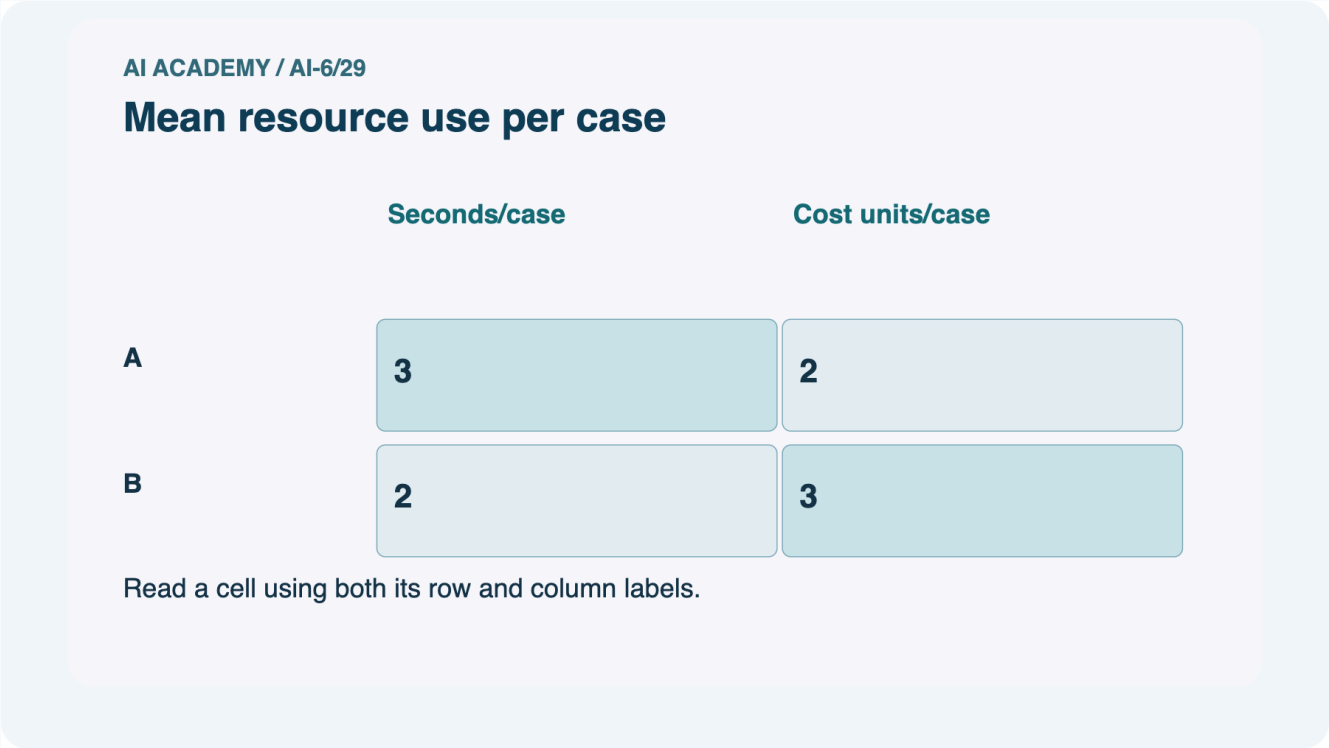
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute mean time: A averages 3 seconds; B averages 2 seconds per case. → Compute mean cost: A averages 2 units; B averages 3 units per case. → Compare trade-off: B saves one second but costs one extra unit per case.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Mean resource use per case



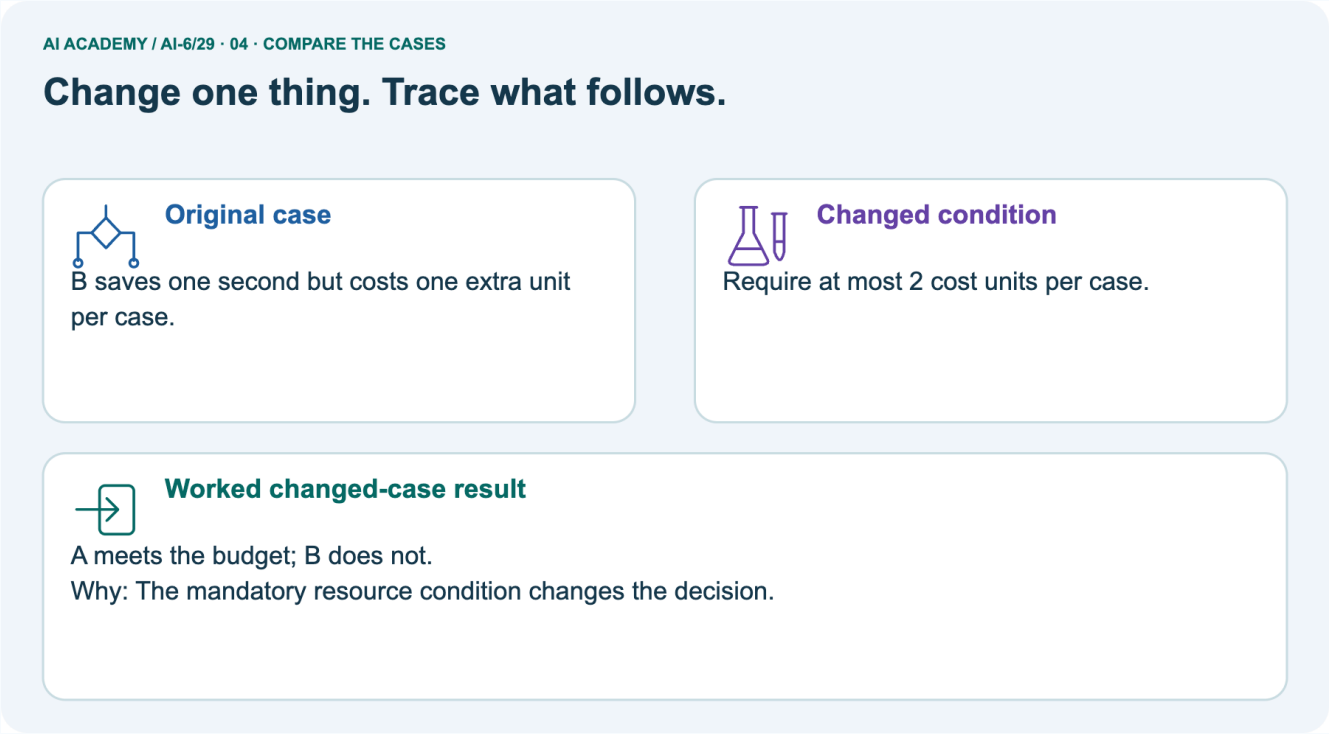
Follow the labelled picture with the matching explanation. Open the vector diagram

Mean resource use per case. A: Seconds/case = 3, Cost units/case = 2. B: Seconds/case = 2, Cost units/case = 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

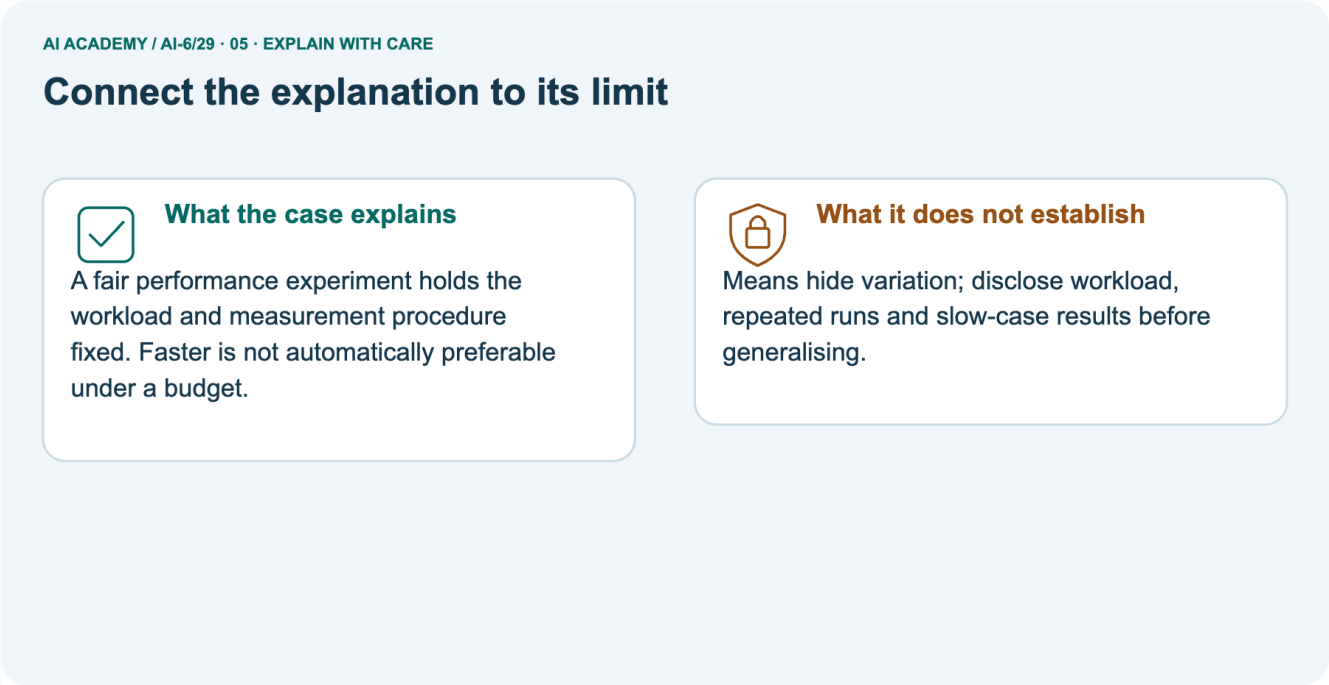
Original case: B saves one second but costs one extra unit per case. Changed condition: Require at most 2 cost units per case. Expected result: A meets the budget; B does not.

Reason: The mandatory resource condition changes the decision.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

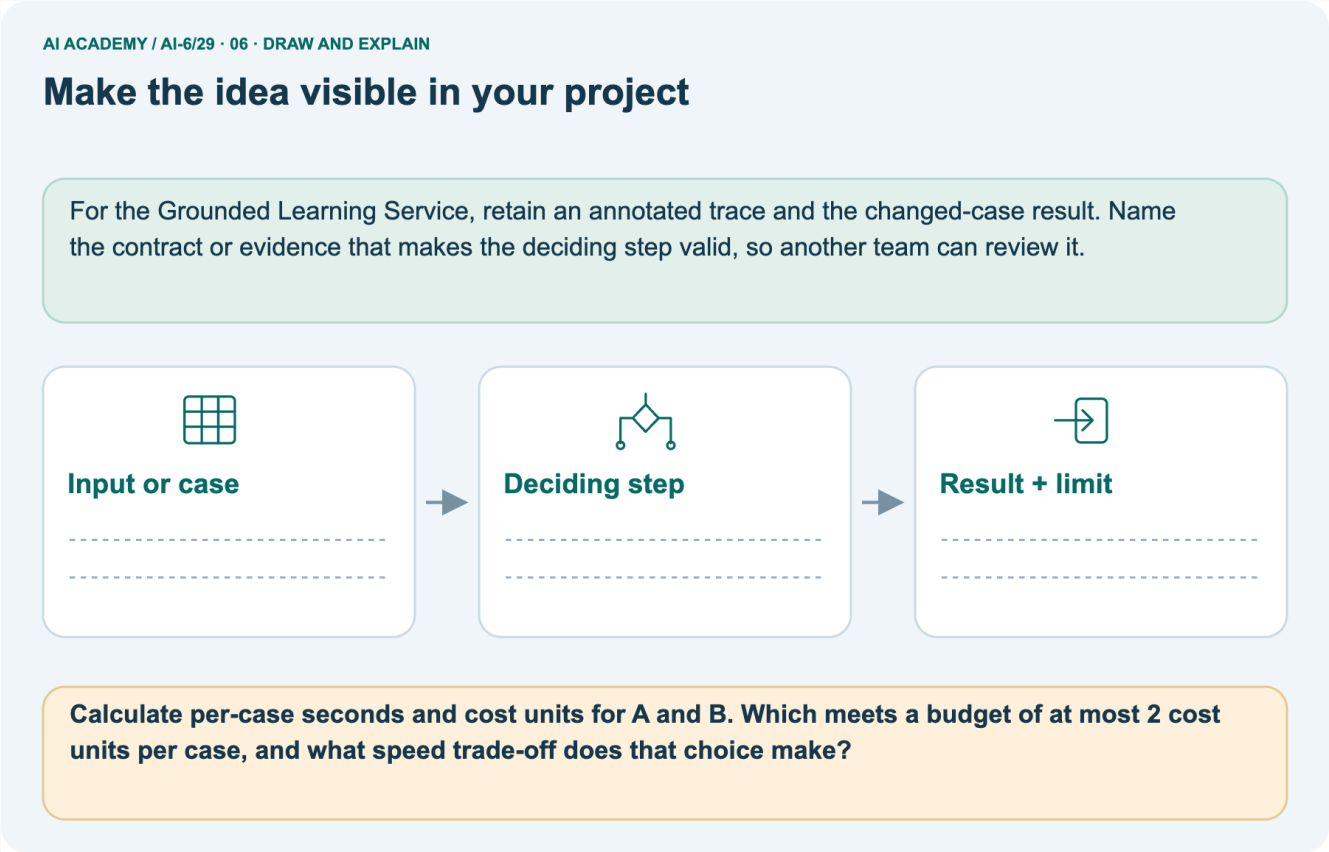
A fair performance experiment holds the workload and measurement procedure fixed. Faster is not automatically preferable under a budget.

Limit: Means hide variation; disclose workload, repeated runs and slow-case results before generalising.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.